- Ass 3 due on 9 October @ 4pm

Recap:

(1) When exact pivots cannot be constructed (e.g., with discrete data) often we can appeal to limit theorems to construct asymptotic pivots

(2) A Bayesian interpretation of probability is that of a (personal) level of belief about unknown parameter(s).

(3) If we have some prior belief $f(\theta)$ about a parameter θ before carrying out an experiment, then after observing data $\underset{\sim}{X} = \underset{\sim}{x}$ from the experiment, we can update our belief to a posterior belief via Bayes' rule:

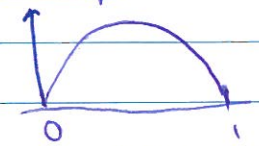
posterior ↘ ↙ prior

$$f(\theta \mid \underset{\sim}{x}) \propto f(\underset{\sim}{x} \mid \theta) f(\theta)$$

↖ likelihood

Example 1 (continued).

- Max's prior: "around 50%"

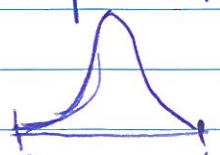


$f(p) \propto p^{2-1}(1-p)^{2-1}$

data: $x = 23$ successes out of $n = 47$

$f(x \mid p) \propto p^{23}(1-p)^{24}$

Max's posterior:



$f(p \mid x) \propto p^{25-1}(1-p)^{25-1}$

$\sim \text{Beta}(25, 25)$

- Jack's prior: "no opinion"

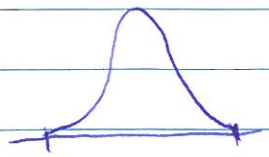


$f(p) \propto p^{1-1}(1-p)^{1-1}$

data: $x = 23$, $n = 47$

$f(x \mid p) \propto p^{23}(1-p)^{24}$

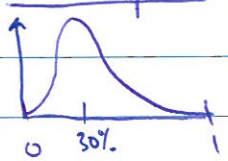
Jack's posterior



$f(p \mid x) \propto p^{24-1}(1-p)^{25-1}$

$\sim \text{Beta}(24, 25)$

- Alan's prior: "strong opinion around 30%"

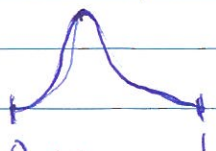


$f(p) \propto p^{30-1}(1-p)^{70-1}$

data $x = 23$, $n = 47$

$f(x \mid p) \propto p^{23}(1-p)^{24}$

Alan's posterior:

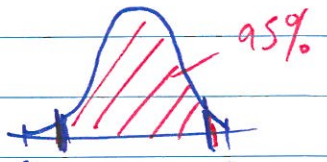


$f(p \mid x) \propto p^{53-1}(1-p)^{94-1}$

$\sim \text{Beta}(53, 94)$

From a Bayesian perspective, the entire posterior distn captures all the information about the parameter after seeing the data. But sometimes it may be useful to summarise our posterior belief via numerical properties. For example, we can report the "middle 95%" of our belief, which quantifies where the "bulk of our belief" lies.

Max's posterior:




```
> qbeta(0.025, 25, 26)        > qbeta(0.975, 25, 26)
[1] 0.355                   [1] 0.626
```

so the "middle 95%" of Max's posterior belief is between (0.355, 0.626)

Jack's posterior:

```
> qbeta(0.025, 24, 25)        > qbeta(0.975, 24, 25)
(0.352                      , 0.628)
```

Alan's posterior

```
> qbeta(0.025, 53, 94)        > qbeta(0.975, 53, 94)
(0.285                      , 0.439)
```

These are called "credible intervals" (CI)

How to interpret these?

For example, Alan might say "I am 95% sure that the true p is between 0.285 & 0.439"

OR even "There is a 95% chance that the true p is between 0.285 & 0.439"

More on this in your tutorial week 8 & STA3500.

For credible intervals, this can be interpreted as a set of parameter values that are consistent with your posterior belief.
For confidence intervals, " " " " " " the data.
In other case, CIs can be thought of as "inverses" of hypothesis tests

§ 7 Hypothesis testing & decision making

Point & interval estimates are important tools for obtaining a "best guess" of an unknown parameter.
However, in applied sciences, we need to make decisions between two competing claims about the world, i.e. H_0 versus H_1, for two claims H_0 & H_1.

A "hypothesis test" or "hypothesis testing procedure" or simply "test" is just a decision rule that specifies when to reject H_0 in favor of H_1.

Example 7.1 A new drug is proposed to lower LDL cholesterol levels in patients at risk of heart disease. To test the efficacy of this new drug, a sample of n (e.g. $n=27$, say) at-risk patients is recruited to a clinical trial. The cholesterol levels of each patient before & after 6 months of taking the drug are recorded, & the differences

$$X_i = \text{before}_i - \text{after}_i \quad , \quad i=1,2,\ldots,n$$

are taken for each patient.

You are asked to make a decision about the effectiveness of this drug based on the data $\underline{X} = (X_1, X_2, \ldots, X_n)^T$

What are some decision rules that **You** can think of for this scenario?

Rule for deciding drug is effective	Underlying pop^n parameter	corresponding hypotheses about parameter
Rule 1: $\min(\underset{\sim}{X}) \geq c_1$ (Harry's rule) for some value c_1 (i.e. $c_1 = 0.3\,\text{mmol/L}$)	pop^n minimum improvement in LDL cholesterol	H_0: min $= 0$ ($\min_0$) vs H_0: min > 0 ($\min_0$)
Rule 2: $M = \text{med}(X) \geq c_2$ for some value c_2 (Jane's rule)	pop^n median, m, improvement	$H_0: m = 0$ (m_0) vs $H_0: m > 0$ (m_0)
Rule 3: $\bar{X} \geq c_3$ for some c_3 (Jack's rule)	pop^n mean, μ, improvement	$H_0: \mu = 0$ vs $H_1: \mu > 0$ (OR $H_0: \mu = \mu_0$, $H_1: \mu > \mu_0$)
Rule 4: $X_7 \geq 1.5$ because I am patient # 7	?	?

There is NO unique decision rule for any given scenario, but intuitively we feel that some rules are more sensible than other rules. How do we formalise our intuition here?

Null hypothesis versus alternative hypothesis

The null & alternative are two competing claims about a pop^n parameter.

As the name suggests, the "null hypothesis" is the "status quo" (or "boring" hypothesis) that essentially says that nothing is happening in our random experiment. The "alternative hypothesis" is the "interesting" hypothesis that claims something is happening in our experiment.

Mathematically speaking, the null & alternative hypotheses are symmetric. However, historically & practically speaking, there are good reasons to treat them asymmetrically (e.g. H_0: innocent versus H_1: guilty are NOT symmetric in the legal system

In this course, we will start with a single point null hypothesis $H_0: \theta = \theta_0$, where θ_0 is a specified null value. The alternative can be composite hypotheses

$H_1: \theta > \theta_0$, or $H_1: \theta < \theta_0$, or two-sided $H_1: \theta \neq \theta_0$.

Example 7.2 In a physics lab experiment, a certain machine is designed to measure acceleration due to gravity, which is currently accepted to be $g = 9.807\ m/s^2$. A lab technician suspects that the machine is miscalibrated. To test this, what would be the null & alternative hypotheses?

$$H_0: \mu = 9.807 \quad \text{vs} \quad H_1: \mu \neq 9.807$$

This is an example of a two-sided alternative.

NOTE: the two hypotheses have NOTHING to do with data

Test statistic

Our decision rule for deciding between H_0 & H_1 needs to be based on the data $\underline{X} = X_1, X_2, \ldots, X_n$ that we will observe.

Of course, we don't just stare at every data point, but rather, we consider an appropriate summary statistic(s) (i.e. a fn of the data). Any such fn of the data can be used as a statistic, & we write

$T = T(X_1, X_2, \ldots, X_n)$ for our test statistic.

Although the choice of statistic is arbitrary, in many scenarios we can use tools/principles to suggest a starting point for our test statistic

For example:

- M.o.M suggest:
  - $\bar{X}$ for testing a pop^n mean μ
  - S^2 for testing a pop^n variance σ^2, & so on...
- alternatively, if we have a model for our data then we can use the MLE as a starting point:

$\bar{X}$ for testing pop^n mean μ if $\underline{X} \sim N(\mu, \sigma^2)$

if $\underline{X} \sim \text{Poisson}(\mu)$

$1/\bar{X}$ for testing pop^n rate λ if $\underline{X} \sim \text{Exp}(\lambda)$

$\max(X_1, X_2, \ldots, X_n)$ for testing θ if $X \sim U(0, \theta)$ (see also Ass 3)

OR we can use a unifying & more powerful approach by appealing to the likelihood ratio principle (next lecture)

Decision rules via critical regions

Once we have identified parameter(s) of interest, formulated our null & alternative hypotheses, & chosen our test statistic $T = T(\underline{X})$, we can then propose a **decision rule** based on the outcome of T:

> Reject H_0 in favor of H_1 if T falls in some critical (rejection) region C

Here, our critical region C can also be one-sided or two-sided reflecting whether the alternative hypothesis is one-sided or two-sided:

right-sided rejection region

$C = [c, \infty)$ i.e., reject $H_0: \theta = \theta_0$ in favor of $H_1: \theta > \theta_0$ if $T \geq c$ (if observed T is "too large")

left-sided rejection region

$C = (-\infty, c]$, i.e., reject $H_0: \theta = \theta_0$ in favor of $H_1: \theta < \theta_0$ if $T \leq c$ (if observed T is "too small")

two-sided rejection region

$C = (-\infty, c_1]$ OR $[c_2, \infty)$ ie, reject $H_0: \theta = \theta_0$ in favor of $H_1: \theta \neq \theta_0$ if T is "too small" OR "too big"

Example 7.1 (continued) For testing mean improvement μ in LDL cholesterol levels (i.e. $H_0: \mu = 0$ vs $H_1: \mu > 0$), one rule that we proposed was

Reject H_0 in favor of H_1 if $\bar{X} \geq 0.3$ mmol/L ⋆

How do we judge the merits of this rule, & how does it compare to competing rules, such as

Reject H_0 in favor of H_1 if $\bar{X} \geq 0.5$ mmol/L ⋆⋆ OR **Reject H_0 in favor of H_1 if $\bar{X} \geq 0.2$ mmol/L** ⋆⋆⋆

⋆⋆: lower type I error but also lower power

⋆⋆⋆: higher type I error, but also higher power

Whenever we make a decision based on some rule, we run the risk of making one of two types of errors:

- accidentally reject H_0 when it is in fact true (Type I)
- accidentally retain H_0 when it is in fact false (Type II)

These two types of errors help us judge the relative merits of any decision rule.

Type I error & significance level of a test

The Type I error is accidentally rejecting H_0 when it is in fact true. The prob. of a Type I error is denoted by

$$\alpha = P(T(\underline{X}) \in C \mid H_0) = P_{H_0}(T(\underline{X}) \in C)$$

("prob. under H_0")

$T(\underline{X}) \in C$: reject H_0; H_0: given H_0 is actually true.

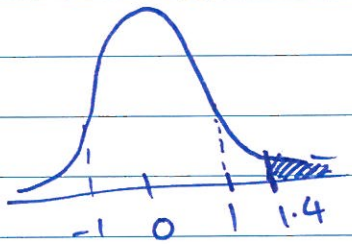
This prob α is called the "significance level" of your test (or just "level").

Computing α requires us to know the dist^n of $T(\underline{X})$ under the null H_0. We can either assume an outright model for our data $\underline{X}$, or we can apply a limit theorem (for example, CLT) for an approx. dist^n of $T(\underline{X})$

Example 7.1 (continued) Consider our medical trial with $n = 27$ patients. A quick search on WebMD suggest that the patient-to-patient variability in cholesterol levels could be as high as $\sigma = 1.1$ mmol/L. Then, what would be the Type I of our decision rule:

Reject H_0 in favor of H_1 if $\bar{X} \geq 0.3$?

$$\alpha = P_{H_0}(\bar{X} \geq 0.3) = P_{H_0}\left(\frac{\bar{X} - 0}{1.1/\sqrt{27}} \geq \frac{0.3 - 0}{1.1/\sqrt{27}}\right)$$



$$\overset{CLT}{\approx} P_{H_0}(Z \geq 1.417)$$

$$= 1 - \text{pnorm}(1.417) = 7.8\%$$

This prob of Type I error is 7.8%.

How about the Type I errors of the two capacity tests:

Reject H_0 if $\bar{X} \geq 0.5$ in favour of H_1

$$\alpha = P_{H_0}(\bar{X} \geq 0.5)$$

$$= P_{H_0}\left(\frac{\bar{X}-0}{1.1/\sqrt{27}} \geq \frac{0.5-0}{1.1/\sqrt{27}}\right)$$

$$\overset{CLT}{\simeq} P_{H_0}(Z \geq 2.362)$$

$$= 1 - \text{pnorm}(2.362)$$

$$= 0.009$$

$$\simeq 1\%$$

Reject H_0 if $\bar{X} \geq 0.2$ in favour of H_1

$$\alpha = P_{H_0}(\bar{X} \geq 0.2)$$

$$= P_{H_0}\left(\frac{\bar{X}-0}{1.1/\sqrt{27}} \geq \frac{0.2-0}{1.1/\sqrt{27}}\right)$$

$$\overset{CLT}{\simeq} P_{H_0}(Z \geq 0.945)$$

$$= 1 - \text{pnorm}(0.945)$$

$$= 17.2\%$$

We see that the weaker the evidence we require, the higher the Type I error (& vice versa).

Suppose we want to control our Type I error at 5% (say). How do we set our critical value in our test?